Project Documentation — Base-Paper Verified Addendum

Companion to Brain_Tumor_MRI_Classification_Project_Report.pdf (five-backbone CNN–LSTM–PCA–SVM framework, explainability analysis, and dataset-integrity audit).

Base paper under verification. Bérété M., Echtioui A., Sellami L., Ben Hmida A. A., *Hybrid CNN+LSTM+SVM model for advanced brain tumor classification*, Network Modeling Analysis in Health Informatics and Bioinformatics, vol. 15, article 43, 2026. DOI 10.1007/s13721-025-00718-8. Published online 02 February 2026.

Scope. This is a documentation-verification addendum, not a rewrite of the report. It (a) verifies the report's descriptions of the base paper against the base-paper PDF, (b) reconciles the report against the transfer-package history and the codebase snapshot, and (c) adds a compact traceability matrix and a final assessment intended for viva defence. It does not modify experimental artifacts and does not reopen training.

Preface — authority order used in this addendum

When sources conflicted, the following order was applied:

- **Base paper PDF** (Bérété et al. 2026) — for every claim about *the paper*.
- **Project report PDF** (*Brain_Tumor_MRI_Classification_Project_Report.pdf*) — for the *current locked* project state, results, and methodology, as instructed.
- **Transfer-package files** (*00_README.md* … *18_HOSTILE_REVIEW_CORRECTIONS.md*) — for historical reasoning and to cross-check that the report did not silently overwrite the audit findings.
- **Codebase references** in the transfer package (*09_CODEBASE_GUIDE.md*, tests, and file listings) — for what is actually implemented.

Where a claim is confirmed by only one source, or is genuinely uncertain, that is stated explicitly rather than papered over.

Section A — Verification of the report against the base paper

Only the report's factual claims *about* the base paper are checked here. The report's own methodology and results are checked in Section B.

A.1 Bibliographic identification

The report cites the base paper as "Bérété, Echtioui, Sellami and Ben Hmida, *Hybrid CNN+LSTM+SVM model for advanced brain tumor classification*, Network Modeling Analysis in Health Informatics and Bioinformatics (Springer), 2026" and marks the DOI, volume, issue, and pages as "citation completion required" (Chapter 10, reference [1]).

Verified against the base paper. Journal, authors, article title and year are correct. The completed reference is:

> Bérété M., Echtioui A., Sellami L., Ben Hmida A. A., "Hybrid CNN+LSTM+SVM model for advanced brain tumor classification," *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 15, article 43, 2026. DOI: 10.1007/s13721-025-00718-8. Published online 02 February 2026. Received 9 July 2025; Revised 6 December 2025; Accepted 9 December 2025.

The affiliations are ATMS/ENIS Sfax (Bérété, Sellami), ENSTA Borj Cedria (Echtioui, corresponding), and King Khalid University (Ben Hmida).

Action for the final report. Replace the "[DOI, volume, issue and pages: citation completion required.]" placeholder in Chapter 10, reference [1], with the verified reference above.

A.2 Task, dataset, and class counts

- The report states the paper's task as four-class MRI classification (glioma, meningioma, no tumor, pituitary). **Confirmed** — base paper Section 3.1.
- The report states the dataset as the Masoud Nickparvar Kaggle *Brain Tumor MRI Dataset*, 7,023 images (Training 5,712; Testing 1,311). **Confirmed** — base paper Section 3.1 gives the same totals and the same Kaggle URL.
- The report reproduces the paper's Testing subclass counts (300/306/405/300). **Confirmed** — base paper Section 3.1.
- Training subclass counts in the paper are glioma 1,321; meningioma 1,339; no tumor 1,595; pituitary 1,457. The report does not repeat these; nothing in it contradicts them.

A.3 Model family, LSTM description, and preprocessing

- The report describes the paper's design as five hybrid pipelines CNN → LSTM → SVM with StandardScaler and PCA between LSTM features and the linear SVM (Chapter 8.3). **Confirmed** — base paper Section 3.2 and Fig. 2.
- The report notes the paper "sometimes phrased" the LSTM as modelling "dependencies across MRI slices". The paper's actual language is *"sequential dependencies across the feature maps"* and "spatial and long-range dependencies", not literal slice or time processing. The "MRI slices" gloss is a project-side paraphrase, not a direct quote from the paper.
- *Correction candidate:* replace "sometimes phrased in terms of dependencies across MRI slices" with "phrased in terms of sequential and long-range dependencies across CNN feature maps".
- The report says the paper used horizontal/vertical flips with brightness ±0.1, contrast/saturation 0.8–1.2, and hue ±0.1. **Confirmed** — base paper Algorithm 1 lists exactly these settings.

A.4 Optimizer matrix, batch size, learning rate, epochs

The report Chapter 8.3 states:

Aspect	Report's description of base paper	Verified?
Optimizer study	5 backbones × 4 optimizers (Adam, SGD, AdaGrad, RMSprop)	Yes — Table 2, §4.2
Batch size	32	Yes — Table 2
Epoch budget	30	Yes — Table 2
Learning rate	1e-4 (report §4.2)	Yes — Table 2
Loss	Categorical cross-entropy	Yes — Table 2
Dropout	0.3	Yes — Table 2
SVM probability	Not stated in report	Paper §3.2: "Probability estimation was enabled
PCA components	Not stated in report	Paper §3.2: min(samples, features, 64) → PCA(
Freezing policy	"Not specified" in the report	Correct — paper does not describe freezing

No incorrect claim was found in the report's description of the paper's protocol. The one refinement worth making is that the paper explicitly enabled probability estimation on its SVM — a difference from the project's current *probability=False* source — and this could be added to Chapter 8.3.

A.5 Best result claimed by the paper

• The report Chapter 8.3 does not restate a numeric headline from the paper; it says the project makes no claim to "match, exceed or explain the paper's reported accuracy". This is correct and defensible.

• For traceability: the base paper's Table 3 and Table 4 report **ResNet152+LSTM+SVM with RMSprop = 99% accuracy, 99% precision, 99% recall, 99% F1**. Adam produces the same 99% top-line for ResNet50 and ResNet152. The paper's Table 4 places four of the five backbones at 99% and EfficientNetB3 at 98% under RMSprop.

A.6 Explainability, integrity audit, GUI in the paper

The report treats Grad-CAM, occlusion, the Streamlit application, the duplicate audit, and the leakage-corrected evaluation as **project extensions** (Chapter 8.3 "Extensions" row). **Confirmed** — the base paper contains no Grad-CAM, no occlusion, no interpretability method of its own, no leakage/duplicate audit, and no GUI. Section 4.4 of the paper mentions future integration of XAI ("Grad-CAM or Score-CAM") as an aspiration, not as work performed.

Verdict for Section A. The report's description of the base paper is substantively accurate. Two small clean-up items are advisable: (i) replace the "MRI slices" gloss with the paper's actual "feature-map sequential dependencies" language; (ii) add one line noting the paper enabled SVM probability estimation.

Section B — Verification of the report against the codebase and transfer package

The report is the authoritative statement of *current* project state. The transfer package and codebase snapshot are read here only as cross-checks.

B.1 Backbones, training policy, and freezing

• Report: five backbones (VGG16, VGG19, ResNet50, ResNet152, EfficientNet-B3); VGG16/VGG19 frozen; the other three fine-tuned. **Cross-check:** *09_CODEBASE_GUIDE.md* states *SUPPORTED_BACKBONES* in *model_cnn_lstm.py* matches exactly, and *train_backbone.py* freezes VGG convolutional blocks. Consistent.

• DenseNet121 is not claimed as an implemented backbone anywhere in the report. Correct — the transfer package's *18_HOSTILE_REVIEW_CORRECTIONS.md* flagged a pre-audit narrative mentioning DenseNet121; the supplied source/manifest/results contain only the five listed backbones. The report aligns with the codebase here.

B.2 Pipeline shape and order

• Report §3, §4 describe: CNN → spatial sequence → LSTM(256) → BatchNorm → Dropout(0.3) → Dense(128) → ReLU (deep feature) → StandardScaler → PCA(64) → linear SVM (C=1, probability=False).

• Cross-check with *09_CODEBASE_GUIDE.md*: matches the described *evaluate_backbone.py* pipeline. The scaler-before-PCA order is verified from code, even though some early paper/project diagrams placed PCA before StandardScaler.

• The report §4.5 statement that the standardized experiments use *probability=False* is consistent with current source. However, the transfer package (*04_DECISION_HISTORY.md* D7, *12_PROBLEMS_DEBUGGING_AND_LESSONS.md*) records that preserved training logs for VGG19 and ResNet152 contain scikit-learn's *probability=True* deprecation warning. This unresolved contradiction is not mentioned in the report.

• *Recommendation:* add one honest sentence in §4.5 or Appendix A.1: "Older log files retain evidence that some legacy runtime SVMs used probability=True; the standardized experiments and the deployed GUI use probability=False with softmax over decision_function() values."

B.3 Data split, seed, and preprocessing

• Report §3.3: 4,569 train, 1,143 validation, 1,311 test; seed 42; stratified 80/20 split of Training; saved indices; direct resize to 224×224; ImageNet normalization; no skull stripping; no segmentation. Cross-check with *07_CURRENT_PROJECT_STATE.md* and *03_PROJECT_JOURNEY_TIMELINE.md*: identical.

• Augmentation: report §4.1 lists *ColorJitter(brightness 0.1, contrast 0.2, saturation 0.2, hue 0.1)*. The base paper uses "contrast 0.8–1.2" and "saturation 0.8–1.2" — the same *effective* range as *ColorJitter=0.2*, because torchvision samples factors uniformly in [1–0.2, 1+0.2] = [0.8, 1.2]. The two specifications therefore match in effect.

• The transfer package repeatedly notes *src/data_pipeline_torch.py* was not in the earlier 25-file attachment. The report's §4.1 asserts a specific transform. The report is authoritative per user instructions; the transform is worth re-checking once before the viva.

B.4 Training-run figures

Report §6.1 per-backbone epoch/time figures were cross-checked with *06_EXPERIMENT_HISTORY.md* Stage 3 and *11_RESULTS_EVOLUTION.md* Stage 3:

Field	Report §6.1	Transfer package	Match?
Best epoch (1-based)	16 / 18 / 7 / 8 / 6	same	Yes
Best NN val acc %	94.58 / 94.58 / 97.73 / 98.25 / 98.43	same	Yes
SVM val acc %	94.93 / 94.49 / 97.55 / 98.25 / 98.51	same	Yes
Original test acc %	94.97 / 93.52 / 98.09 / 98.40 / 98.47	same	Yes
Original macro F1 %	94.68 / 93.10 / 98.01 / 98.32 / 98.41	same	Yes

The report is self-consistent and consistent with the transfer package. The "zero-based vs one-based epoch" convention flagged in the transfer package is resolved in the report by an explicit note in §6.1 ("Best epochs are 1-based").

B.5 Duplicate audit and leakage-corrected evaluation

- Report §6.3 gives 464 exact + 98 high-confidence = 562 material cross-split pairs, 481 review-only, verdict *MATERIAL_DUPLICATION*, dated 20 Sep 2026 17:53 UTC. *07_CURRENT_PROJECT_STATE.md*, *06_EXPERIMENT_HISTORY.md* §E6, and the appendix excerpt in the report agree exactly.
- Report §6.4 gives three exclusion policies (exact_clean 1,095; material_clean 1,061 [primary]; conservative_clean 926) with per-class counts, and states no CNN was retrained and no scaler/PCA/SVM refitted. Consistent with the "post-hoc no-retraining correction" described in §3.5.
- The primary corrected accuracies (EfficientNet-B3 98.680%; ResNet152 98.398%; ResNet50 98.303%; VGG16 94.816%; VGG19 93.025%) are the numbers the user has designated as the project's final, locked classification results.

B.6 Explainability results

- Report §6.5 gives the Grad-CAM ratings (L3 = 1/9/5; L4 = 3/2/10; Fused = 2/6/7 across 15 tumor images), the shared error on *Te-pi_0078.jpg*, and the occlusion counts (1,859 patches, 58 flips, 32×32 patch, stride 16, 169 per image).
- Cross-check: identical to transfer-package *06_EXPERIMENT_HISTORY.md* §E4/§E5 and to *11_RESULTS_EVOLUTION.md*. Consistent.

B.7 Positioning against pre-audit ("not locked") language

The transfer package explicitly framed the project as **not** ready to lock because a duplicate-group-aware retraining was pending (*00_README.md* "Critical status warning", *07_CURRENT_PROJECT_STATE.md*, *08_FINAL_METHODOLOGY.md*).

The user's later checkpoint (recorded in project memory) supersedes that: the experimental phase is complete and locked, the corrected evaluation is sufficient for the academic project, and the group-aware rerun is future work. The report §6.4 and §9 adopt exactly this stance, with the explicit disclaimer that results are "leakage-corrected at the identified test-image level" rather than "completely leakage-free". **This is the correct wording for the current project scope** and is not overreach; it is also clearly labelled as future work.

PART 17 — Base-Paper Reproduction Traceability Matrix

Rows are the elements the base paper defines; columns state where each is implemented in the project, evidence file(s), and what differs. "Report §X" refers to *Brain_Tumor_MRI_Classification_Project_Report.pdf*.

#	Base-paper element	Paper	Implemented?	Project file / evidence	Difference / note
1	Four-class MRI classification (glioma / mening. / notumor / pituitary)	§3.1	Yes	Report §1, §3.3; 04_DECISION_HISTORY D2	Same task; identical class names and label ordering
2	Dataset: Kaggle Nickparvar Brain Tumor MRI (7,023; 5,712+1,311)	§3.1	Yes	Report §1, §3.3; Report Appendix A	Same dataset; audit later showed pre-existing cross-split duplication in the supplied partitions
3	Test-set class counts (300 / 306 / 405 / 300)	§3.1	Yes	Report §3.3, §6.3	Identical
4	Data augmentation (H/V flip; bright ±0.1; contrast 0.8–1.2; sat 0.8–1.2; hue ±0.1)	Alg. 1	Yes (equivalent)	Report §4.1; src/data_pipeline_torch.py	torchvision ColorJitter(0.1/0.2/0.2/0.1) yields the same ranges
5	Framework: TensorFlow	§3.2	**No — deliberate**	Report §1; 04_DECISION_HISTORY D4	Migrated to PyTorch after RTX 50-series sm_120 PTX incompatibility; environmental, not scientific
6	Backbones: VGG16, VGG19, ResNet50, ResNet152, EfficientNetB3	§3.2	Yes, all five	Report §4.3; model_cnn_lstm SUPPORTED_BACKBONES	Same five; independent alternatives, not an ensemble
7	ImageNet pretraining of CNN backbones	§3.2	Yes	Report §4.3 (V1/V2 weight strings)	Same
8	CNN → reshape → LSTM chain	§3.2, Fig. 2	Yes	Report §3.2, §4.2	Report clarifies the LSTM sees spatial positions, not time — a valid rewording
9	LSTM with forget/input/output gates (Eq. 1–6)	§3.2	Yes (nn.LSTM)	Report §4.2	Hidden 256, batch-first, one layer
10	BatchNorm + Dropout + Dense after LSTM	§3.2, Fig. 2	Yes	Report §3.2, §4.2	BatchNorm1d → Dropout(0.3) → Linear(256→128) → ReLU
11	StandardScaler (zero mean, unit variance)	§3.2	Yes	Report §4.5; evaluate_backbone.py	Fitted on training-only, as the paper prescribes
12	PCA to min(n_samples, n_features, 64) — effectively PCA(64)	§3.2	Yes	Report §4.5	Same; explained variance 99.18–99.45% per backbone
13	Linear SVM, C = 1.0	§3.2 Eq. 7–8	Yes	Report §4.5	Same kernel and regularization
14	SVM probability=True	§3.2	**No — differs**	Report §4.5; 12_PROBLEMS_DEBUGGING_AND_LESSONS	GUI uses decision_function(); preserved logs show True on some legacy runs
15	Optimizer study: Adam + SGD + AdaGrad + RMSprop (5×4 = 20 runs)	§4.2, Tab. 2	**Partial — RMSprop only**	Report §8.3, §9.3 future work	~4× less compute than the paper
16	Learning rate 1e-4	Tab. 2	Yes	Report §4.4	Same
17	Loss: categorical cross-entropy	Tab. 2	Yes	Report §4.4	Same
18	Batch size 32	Tab. 2	**No — differs** (8)	Report §4.4	Small-batch training with drop_last=True for BatchNorm1d
19	Max epochs 30	Tab. 2	**Partial**	Report §4.4, §6.1	10 default with early stopping, VGGs extended to 30
20	Dropout 0.3	Tab. 2	Yes	Report §4.4	Same
21	Freezing / fine-tuning policy	Not spec.	Project-specific	Report §4.3; 06_EXPERIMENT_HISTORY F2	VGG16/VGG19 frozen; others fine-tuned — asymmetry not in the paper

#	Base-paper element	Paper	Implemented?	Project file / evidence	Difference / note
22	Metrics: accuracy, precision, recall, F1	§4.1	Yes	Report §6.2, §6.4	Same; also per-class support and macro/weighted F1
23	Confusion matrices (per optimizer × backbone)	Fig. 5–8	Partial (RMSprop only)	Report §6.2, §6.4, Fig. 6.4	Report shows ResNet152 and EfficientNet-B3 corrected matrices
24	Comparative table vs. other state-of-the-art	Tab. 4	Not attempted	Report §8.4	Report does not claim to outperform paper's cited SoTA
25	Best result (ResNet152+LSTM+SVM RMSprop = 99% acc/F1)	Abs., Tab. 3, 4	Not reproduced at same headline	Report §6.2, §6.4	Project ResNet152 = 98.40% orig, 98.398% corr; ~0.6 pp below paper. See §18 below
26	Grad-CAM / other XAI	Absent (future work in §4.4)	**Yes — project extension**	Report §4.6, §6.5	Grad-CAM on the neural head at ResNet layer 3, 4, and fused
27	Occlusion sensitivity on the full pipeline (final SVM)	Absent	**Yes — project extension**	Report §4.6, §6.5, Fig. 6.5	Interrogates the deployed classifier where Grad-CAM cannot
28	Streamlit / interactive research GUI	Absent	**Yes — project extension**	Report §4.8; app.py, gui/*	Four-tab research app
29	Cross-split duplicate / near-duplicate audit	Absent	**Yes — project extension**	Report §4.7, §6.3	Hashing + pHash + dHash + MAE/correlation across three partition pairs
30	Leakage-corrected evaluation (post-hoc, no retraining)	Absent	**Yes — project extension**	Report §3.5, §6.4	Primary policy material_clean (n=1,061); two sensitivity policies
31	Statistical rigour (multiple seeds, CIs, paired tests)	Absent	Not implemented	Report §7.2, §9.3	Both project and paper are single-seed/single-split
32	Ablations (no-LSTM, no-PCA, alternative orderings)	Absent	Not implemented	Report §7.2, §9.3	Future work in both
33	External-dataset / patient-level validation	§4.4 as future	Not implemented	Report §7.2, §9.3	Future work in both

PART 18 — Final Assessment

The following answers are evidence-based, use only material verified in this addendum, and follow the user's instruction not to flatter the project.

18.1 Did we successfully implement the core architecture proposed by the base paper?

Yes. All five backbones are implemented in the same CNN → spatial-sequence → LSTM → BatchNorm → Dropout → Dense(128) → StandardScaler → PCA(64) → linear SVM pipeline the paper describes (traceability rows 6–13). The five pipelines run end-to-end, produce metrics, and drive the Streamlit application. This part of the reproduction is complete.

18.2 How closely did we reproduce the methodology?

Partially, with disclosed deviations. The reproduction is faithful in architecture, loss, learning rate, dropout, PCA size, and SVM kernel/C. It diverges deliberately in:

- **Framework** — PyTorch instead of TensorFlow (RTX 50-series sm_120 incompatibility; environmental, not scientific).
- **Optimizer scope** — RMSprop only, not the paper's 5×4 matrix. The project explored roughly one-fifth of the paper's experimental grid.
- **Batch size** — 8, not 32; a hardware/practical choice that matters because BatchNorm1d must drop the last batch of one image (drop_last=True).
- **Epoch budget** — 10 default with early stopping, VGGs extended to 30; the paper uses 30 fixed.
- **VGG backbone freezing** — the paper describes no freezing; the project froze VGG16 and VGG19 after training collapse and fine-tuned the other three. Non-trivial methodological asymmetry, correctly flagged by report Chapter 6.6.
- **SVM probability flag** — paper enables it; the project uses probability=False, with the GUI using decision_function().

None of these deviations is hidden in the report; each is documented in §5, §4.3, and §8.3. The strongest single-line honest description is **"paper-inspired, not an exact reproduction"** — which is exactly what the report Chapter 1 says.

18.3 How closely did our results reproduce the paper's reported performance?

The paper's headline is **ResNet152+LSTM+SVM (RMSprop) = 99% accuracy / 99% precision / 99% recall / 99% F1** (Bérété et al. Table 3, Table 4). Under RMSprop the paper reports four backbones at 99% and EfficientNetB3 at 98%. The project's results are:

Backbone	Paper acc (RMSprop) %	Project original test acc %	Project corrected (material_clean) %
VGG16	94	94.97	94.816
VGG19	96	93.52	93.025
ResNet50	99	98.09	98.303
ResNet152	99	98.40	98.398
EfficientNet-B3	98	98.47	98.680

Interpretation. VGG16 and EfficientNet-B3 land within ~1 pp of the paper, in both directions. ResNet50 and ResNet152 land ~0.6–0.9 pp below the paper's 99%. VGG19 lands ~3 pp below the paper — the largest gap and the only one worth calling out. Given the project used a different batch size, a different (shorter) epoch budget for the non-VGG backbones, and a frozen VGG policy that the paper does not describe, near-identical numbers for three backbones and one meaningful VGG19 gap is a **credible partial reproduction**, not a headline-matching one. The report is correctly cautious and does not claim to

replicate the 99% figure.

18.4 What important differences exist?

Beyond the training-config differences listed in 18.2:

- **Freezing asymmetry.** The five-backbone comparison is not "architecture only"; VGGs are fixed feature extractors with a trained read-out while the other three adapt their convolutional weights. The report §4.3 states this.
- **Data-integrity finding.** The project audit found pre-existing cross-split duplication in the paper's own dataset partition (464 exact + 98 high-confidence = 562 material cross-split pairs). This affects anyone who uses those directories as-is — including the base paper, whose results are reported without any duplicate check. Arguably the single most consequential difference between the two works.
- **Explainability and integrity integrated into the study.** The base paper's §4.4 treats XAI and larger validation as future work; the project performs them at small scale with disclosed caveats and folds their findings back into the main interpretation (report §6.5, §6.6). A methodological difference in kind, not just scope.

18.5 What did we add that the base paper did not contain?

Concretely (traceability rows 26–30):

- **Grad-CAM on the differentiable neural head**, with a fixed 20-image protocol at seed 42 and manual ratings.
- **Full-pipeline final-SVM occlusion sensitivity** with prediction-flip counts and normalization caveats. Important because Grad-CAM cannot back-propagate through the deployed scikit-learn SVM.
- **Cross-split duplicate/near-duplicate audit** using five complementary measures (SHA-256, pixel SHA-256, pHash, dHash, MAE + correlation).
- **Post-hoc leakage-corrected evaluation** under three predefined policies, with the primary policy chosen before results were examined.
- **Streamlit research application** loading standardized per-backbone artifacts, with a clear separation of neural-head and SVM outputs, and a research-warning disclaimer.

None of the above appears in the base paper.

18.6 Are Grad-CAM and final-SVM occlusion a defensible mini-project-level extension?

Yes, at the mini-project / undergraduate research level. Grad-CAM is a well-established method (Selvaraju et al. 2017), and its use on hybrid CNN-based classifiers is standard. Where the project earns its credibility is in *not* overclaiming: the report makes three points that a hostile reviewer will look for and finds satisfied — (i) Grad-CAM explains the neural head, not the deployed SVM; (ii) occlusion is added specifically because Grad-CAM cannot explain the SVM; (iii) occlusion maps are normalised per image so bright red does not mean "large effect", and the flip counts are reported alongside the maps. The single-rater, 20-case protocol is a legitimate weakness (report §6.5, §7.2) — but it is stated openly. If a reviewer objects that occlusion is not a novel method, the correct reply is that novelty was never claimed; the extension is a *methodological* one — matching each explainer to the classifier it can actually explain — not an algorithmic one.

18.7 Does the leakage audit materially change the project conclusion?

Yes, for interpretation; no, for the numeric ranking.

- For **interpretation**: it downgrades the strength of the headline claim. Before the audit, ~98% on a public benchmark would be presented as a strong within-dataset result. After the audit, the honest

statement is ~98% on a test set that shared pre-existing image-level content with the training partitions, and remained ~98% on the 1,061-image material-clean subset. That is a weaker but truer claim.

• For the **numeric ranking**: the post-hoc corrected accuracies moved by less than 0.5 pp for every backbone, and the ordering (three residual/EfficientNet models above the two frozen VGG models) was preserved. The audit does not overturn the comparison.

• For **future work**: the audit produces a specific, non-speculative next step — duplicate-group-aware retraining — which the project deliberately defers rather than pretends to have done. Disclosed in §6.4 and §9.3.

18.8 What is the strongest defensible contribution of the completed project?

Not the accuracy number. The strongest, most defensible contribution is the **methodological triangulation** it demonstrates on a public MRI benchmark: *performance* × *explainability* × *dataset integrity, examined together on the same trained pipelines.* Concretely:

• A standardized, artifact-preserving five-backbone comparison that made the leakage audit and corrected re-scoring possible *without* retraining (report §6.4 explicitly relies on the saved test-PCA features and SVMs).

• Two explanation paths, each matched to the classifier it can actually explain, with quantified normalization/effect-size caveats — a discipline the base paper does not exercise.

• A duplicate audit that identified pre-existing cross-split contamination in a widely used public dataset and quantified its effect. Whether or not any reviewer accepts the specific pHash/dHash thresholds, the finding is reproducible and independently checkable.

That combination — not the ~98% — is what a viva committee should be directed toward.

Appendix — Brief audit summary

Overall reproduction classification

Paper-inspired partial reproduction with disclosed deviations, plus three project extensions (Grad-CAM / SVM-occlusion / integrity audit + correction) that the base paper does not contain.

Most important matches with the base paper

Architecture family (5 CNN backbones × LSTM × PCA × linear SVM); dataset (same Kaggle 7,023-image partitioning); pre-processing order (StandardScaler before PCA, both fit on training only); ImageNet-normalised 224×224 input; class scheme (glioma, meningioma, no tumor, pituitary); RMSprop optimizer choice; learning rate 1e-4; dropout 0.3; PCA 64 components; linear SVM C = 1.0.

Most important deviations from the base paper

Framework (PyTorch instead of TensorFlow); optimizer coverage (RMSprop only, not the 4-optimizer × 5-backbone matrix); batch size 8 versus 32; epoch budget (10 with early stopping, 30 for VGGs; paper uses 30 fixed); frozen VGG16/VGG19 convolutional backbones (paper does not describe freezing); SVM *probability=False* in current standardized source (paper enables probability estimation); use of GUI's *decision_function()* softmax as "relative class scores".

Paper result vs. project result (headline)

Paper: ResNet152+LSTM+SVM RMSprop = 99% accuracy / 99% F1. Project: ResNet152 = 98.40% (original 1,311 test set) → 98.398% (material-clean 1,061 test subset). Top backbone in this project is EfficientNet-B3 at 98.680% (corrected); the paper's own top backbone (ResNet152) reached 99% in the paper but 98.398% here. **We do not match the 99% headline; we are consistently within ~1 pp for four of the five backbones and ~3 pp below on VGG19.**

Does the base paper contain explainability?

No. The paper mentions Grad-CAM/Score-CAM only in its Discussion as future work. It reports no saliency, no occlusion, no faithful-explanation study, and no interpretation tool.

Is our Grad-CAM / occlusion work an extension?

Yes. Both are added by this project. Occlusion in particular was added because Grad-CAM cannot back-propagate through the scikit-learn SVM.

Is the leakage audit an extension?

Yes. The base paper does not audit duplicates and does not report any leakage-corrected evaluation. Both are project extensions, and both are performed on the paper's own dataset partitioning.

Claims in the report that were corrected in this addendum

- Base-paper reference in Chapter 10 [1] previously left DOI/volume/pages as "citation completion required"; the verified citation is now supplied above (DOI 10.1007/s13721-025-00718-8, vol. 15, article 43).
- The gloss "sometimes phrased in terms of dependencies across MRI slices" in Chapter 8.3 is a project rephrasing; the paper's actual language is "sequential dependencies across the feature maps". Wording adjustment suggested.
- Chapter 4.5 says the standardized experiments use *probability=False*, which is correct. The paper enables SVM probability estimation; that contrast is worth adding as a one-line note.
- The augmentation numbers in report §4.1 (*ColorJitter(0.1, 0.2, 0.2, 0.1)*) are the same effective range as the paper's "0.8–1.2" for contrast and saturation; if a reviewer asks, the equivalence is exact (torchvision samples factors in [1–x, 1+x]).

Base-paper details that remain genuinely ambiguous

• The paper does not state a CNN-backbone freezing/fine-tuning policy; how it stably fine-tuned VGG16 in the same 30-epoch RMSprop regime that caused this project's VGG16 collapse is unexplained. The paper's own VGG16 result under RMSprop is 94% — close to this project's frozen VGG16 at 94.8%.

• The paper does not specify a random seed, train/validation split policy, or number of independent runs. Its near-identical 99% headlines are therefore not directly comparable to single-seed single-split project results.

• The paper does not report any duplicate/leakage check on the Kaggle Nickparvar dataset, so the effect of the pre-existing cross-partition duplication on its own headline numbers is unknown.

End of Base-Paper Verified Addendum.